

WEEK 7 PROJECT SUMMARY

Data Analytics Track | Testing, Refinement & End-to-End Validation

WEEK 7 AT A GLANCE

Week 7 continued directly from the Week 6 no-show risk analysis. The work focused on testing the documented KPI and analytical findings, reviewing dashboard consistency and interpretation, refining conclusions to avoid unsupported causal claims, and defining the remaining validation required before Week 8. The strongest documented indicators remain previous no-show history and booking lead time. Cross-track testing remains an outstanding dependency because no collaborator output was received.

1. Planned Testing

Area	Week 7 intention
KPI	Recheck the 48.46% no-show KPI against the documented Week 6 population and numerator.
Analytical findings	Reassess previous no-shows, booking lead time, appointment type and reminder-channel findings.
Dashboard	Duplicated the Week 6 page to preserve the baseline, then refined the Week 7 page with benchmark lines, a trend line, clearer labels and revised observed-rate wording.
Recommendations	Ensure business actions are linked to observed evidence and do not imply causality.
Cross-track	Attempted to connect the findings to Data Science validation requirements; no collaborator output was received, so no completed cross-track test is claimed.

2. What Was Actually Tested

Component	Testing performed	Result
Overall KPI	Rechecked the documented 2,423 no-shows against 5,000 appointments.	48.46%; consistent with Week 6 evidence.
Previous no-shows	Reviewed the documented 0-3 previous-no-show pattern.	Increasing observed rates remain consistent with Week 6.
Booking lead time	Reviewed the five documented booking lead-time bands.	Increasing observed rates remain consistent with Week 6.

Appointment type	Reassessed the four category rates and Week 6 interpretation.	Rates remain relatively close; not a strong standalone targeting basis.
Reminder channel	Reviewed the documented channel rates and interpretation.	SMS remains a testing candidate, not a proven causal intervention.
Dashboard interpretation	Reviewed the KPI, four visuals, Key Takeaway and Business Insight/Action panels.	Structure remains decision-oriented; wording was refined to preserve observational interpretation.

3. Most Important Testing Results

The Week 7 review retained the central Week 6 findings rather than introducing a new analytical story:

- The overall no-show KPI remains 48.46%, representing 2,423 no-shows out of 5,000 appointments.
- Previous no-show history shows an increasing observed no-show rate across the main interpretable groups.
- Booking lead time shows a steady increase in observed no-show rates from 24.18% for 0-2 days to 60.49% for 31+ days.
- Appointment-type rates are relatively close and therefore should not be treated as the primary targeting basis.
- Reminder-channel differences remain observational; SMS is a candidate for structured further testing rather than a proven intervention.

4. Issues / Weaknesses Identified

- The supplied evidence does not include the original CSV dataset, so raw-data recalculation and stability testing on additional data could not be claimed.
- The dashboard PBIX was available, but live execution of every Power BI filter/calculation was not independently performed for this report.
- Observed relationships do not establish causation.
- Reminder-channel comparisons may be affected by patient characteristics, operational processes or targeting decisions.
- Very small previous-no-show groups require caution and were not used as the primary evidence for the main trend.
- The mandatory cross-track testing activity could not be completed because no collaborator/model output was received.

5. Refinements Made

Refinement	Reason
Dashboard visual refinements	Duplicated Week 6 into a dedicated Week 7 page; added a 48.46% benchmark to Appointment Type and Booking Lead Time, an upward trend line and clearer labels to Previous No-Shows, and revised Reminder Channel to “Observed No-Show Rates”.
Interpretation refinement	Kept previous no-shows and booking lead time as the clearest observed indicators while avoiding causal claims.

Reminder-channel refinement	Reworded the visual and insight/action text to present channel differences as observed associations and a testing opportunity.
Validation layer	Added a Week 7 Testing & Refinement header and validation panel to make the progression from Week 6 explicit.
Evidence/status distinctions	Maintained the separation between completed documented checks and outstanding raw-data, predictive and cross-track tests.

6. Retesting / Validation Status

Requirement	Status	Outcome / remaining action
KPI validation	PASS	48.46% agrees with the documented Week 6 calculation.
Analytical pattern review	PASS — documented evidence	Week 6 patterns remain internally consistent.
Dashboard consistency	PARTIAL	PBIX/dashboard evidence reviewed; live interaction testing remains outstanding.
Raw-data validation	NOT COMPLETED	Requires access to the approved HealthConnect appointment dataset.
Additional-data stability	NOT COMPLETED	Requires new/additional data.
Cross-track testing	NOT COMPLETED	Requires a meaningful Test → Finding → Action → Retest activity with another HC-POD track.

7. Cross-Track Testing Status

STATUS: OUTSTANDING

Collaboration was attempted during Week 6/7, but no usable Data Science output was received. No model change, joint test, exchanged dataset or validated cross-track improvement is claimed. The Data Analytics hand-off is nevertheless prepared: previous no-show history and booking lead time are proposed candidate predictors; reminder channel is a secondary candidate requiring confounding/effectiveness testing; and the baseline outcome is the documented 48.46% no-show rate.

8. Key Decisions

- Retain 48.46% as the headline no-show KPI.
- Prioritise previous no-show history and booking lead time as the clearest observed risk indicators.
- Do not use appointment type alone as a targeting variable.
- Do not treat the SMS observation as proof of effectiveness.
- Do not claim cross-track model validation where no collaborator output was received.
- Carry the documented candidate variables and testing questions into the next modelling/validation stage.

9. Contribution to the Overall HealthConnect Project

The Data Analytics contribution provides a tested and documented descriptive evidence base for the wider HealthConnect solution. It identifies candidate risk indicators, links them to operational questions, preserves analytical limitations, and defines the validation questions that a future predictive modelling stage should address.

10. Remaining Issues & Week 8 Requirements

Remaining item	Required before / during Week 8
Cross-track validation	Complete a meaningful testing activity with the relevant HC-POD track and document Test → Finding → Action → Retest → Validated Improvement where applicable.
Raw-data validation	If the approved dataset is available, reconcile key dashboard calculations directly to the underlying records.
Predictive validation	Assess whether previous no-shows and booking lead time add useful predictive information beyond other variables.
Reminder testing	Use appropriate statistical or experimental testing before recommending broader channel changes.
Final integration	Compare validated Data Analytics findings with other track outputs and resolve material differences.

11. Final Week 7 Assessment

Week 7 demonstrates forward progress from the Week 6 deliverable through structured review, refinement of interpretation and clearer separation between validated documented findings and outstanding tests. The Data Analytics work is not presented as fully complete against every Week 7 requirement because the mandatory cross-track testing activity and certain raw-data-level tests remain outstanding.